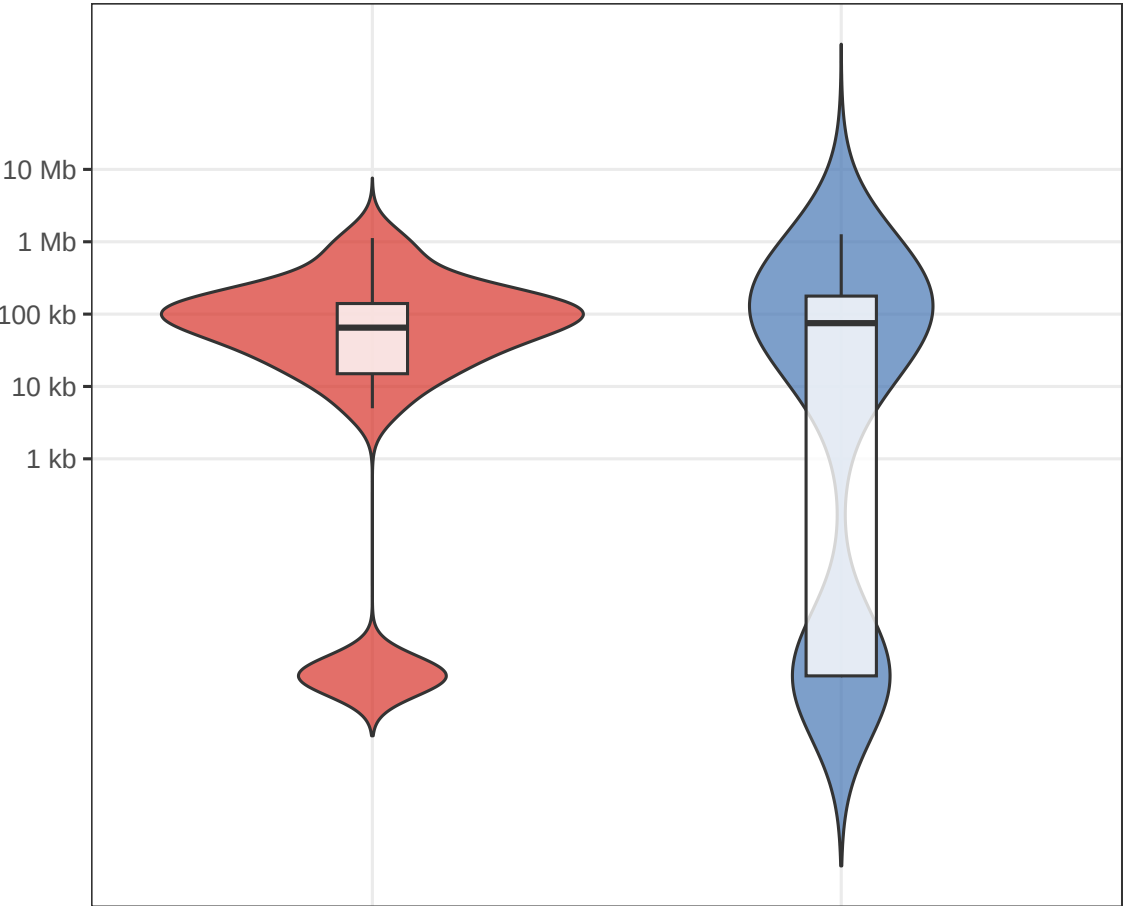


Boundary–Loop Cross–Reference: Early Timepoint

Loop Anchor Distance to Differential TAD Boundaries

Early Timepoint | Wilcoxon $p = 0.704$ | Lost median = 74,999, Gained median = 64,999

Distance to nearest differential boundary



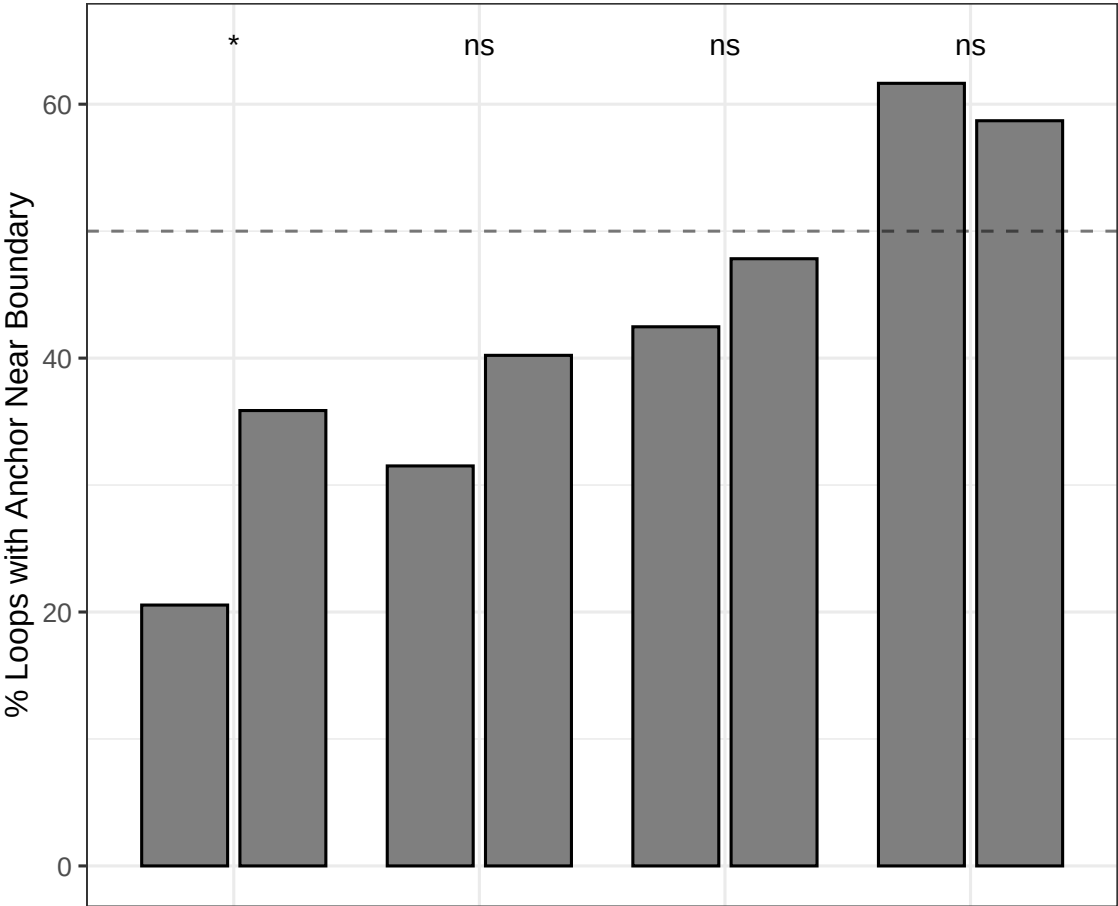
Gained in mutant

Lost in mutant

Loop Direction

Proportion of Loops Near Differential Boundaries

Early Timepoint



10kb

25kb

50kb

100kb

Distance Threshold

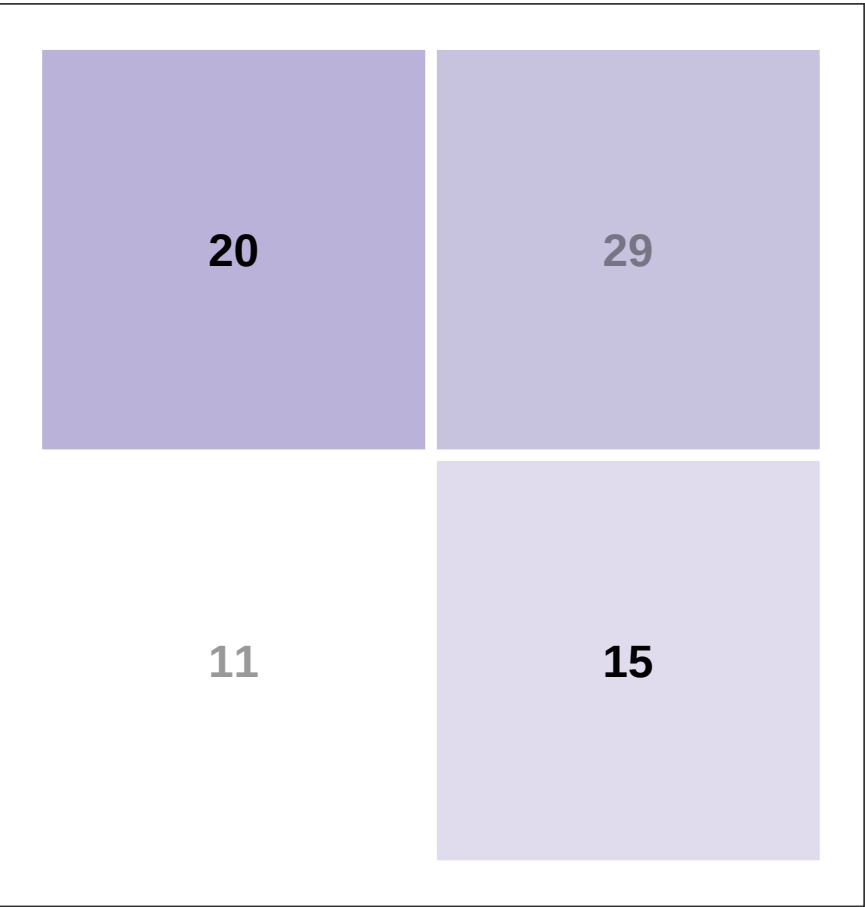
Loop Direction × Boundary Enrichment Concordance

Early Timepoint | Concordance rate: 46.7% ($\chi^2 p = 1$)

Nearest Boundary Enriched In

Mutant-enriched

Control-enriched



Gained

Lost

Loop Direction

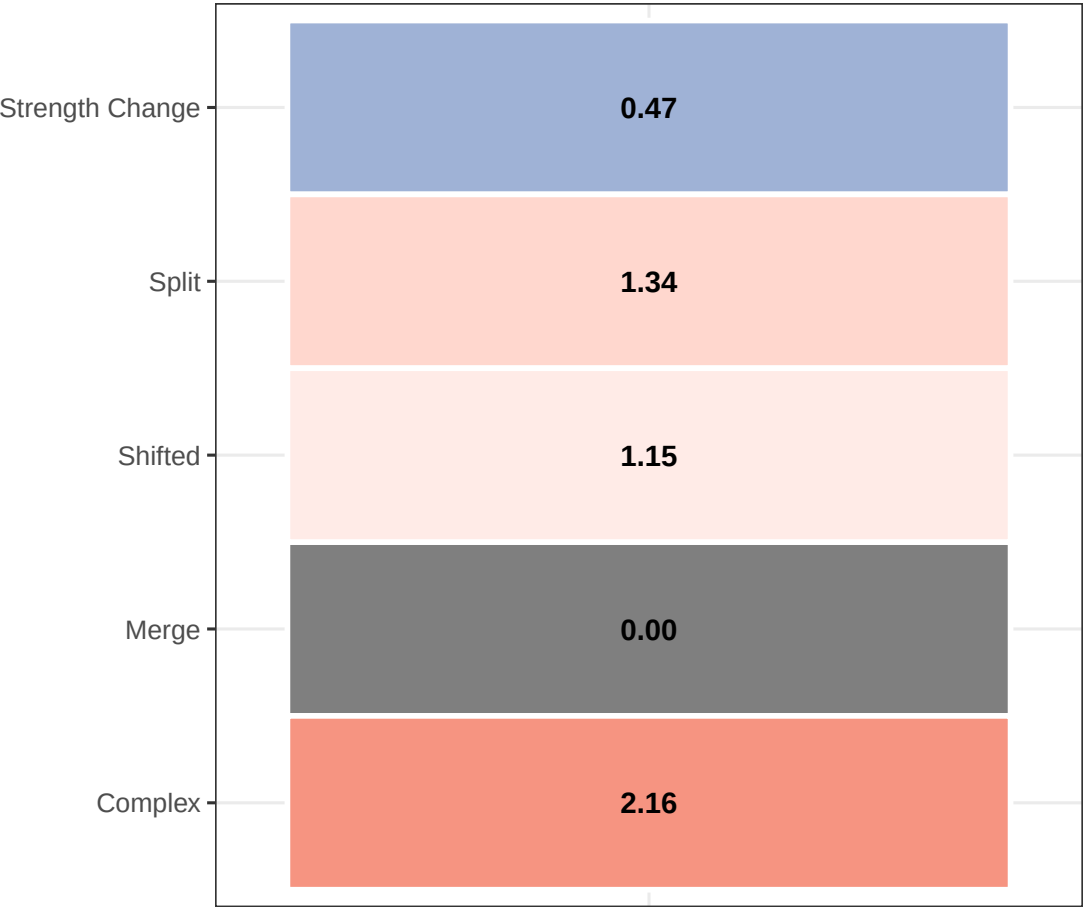
Count

25
20
15

Boundary Type

Boundary Type Association with Loop Direction

Early Timepoint | OR > 1 = enriched in gained loops



Strength Change

Split

Shifted

Merge

Complex

0.47

1.34

1.15

0.00

2.16

log2(OR)

2
1
0
-1
-2